Fractions, Decimals and Percentages

Name _____ Date _____ Calculator allowed Version 1

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KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · convert *(panoni)*

A Converting ●○○

Remember: Per cent means "out of 100". To change a decimal to a percentage, multiply by 100.

1. Write each fraction as a decimal.

a) $\frac{3}{4}$ = _____ **b)** $\frac{2}{5}$ = _____ **c)** $\frac{7}{8}$ = _____ **d)** $\frac{3}{20}$ = _____

e) $\frac{1}{8}$ = _____ **f)** $\frac{9}{40}$ = _____ **g)** $\frac{17}{50}$ = _____ **h)** $\frac{5}{16}$ = _____

2. Write as a percentage.

a) 0.35 = _____ **b)** 0.07 = _____ **c)** 1.2 = _____ **d)** $\frac{3}{5}$ = _____

e) $\frac{7}{25}$ = _____ **f)** $\frac{5}{8}$ = _____ **g)** $\frac{9}{20}$ = _____ **h)** 0.004 = _____

3. Write as a fraction in simplest form.

a) 0.6 = _____ **b)** 0.45 = _____ **c)** 12% = _____ **d)** 37.5% = _____

e) 0.125 = _____ **f)** 0.04 = _____ **g)** 85% = _____ **h)** 2.5% = _____

B Ordering ●●○

Remember: Change every number to the same form, such as decimals, before you compare them.

4. Write each set in order, smallest first.

a) $0.6, \quad 58\%, \quad \frac{5}{8}$

b) $\frac{1}{3}, \quad 0.3, \quad 35\%$

c) $72\%, \quad \frac{3}{4}, \quad 0.7$

d) $1.2, \quad 125\%, \quad \frac{9}{8}$

e) $0.45, \quad 44\%, \quad \frac{4}{9}$

f) $2.1, \quad 205\%, \quad \frac{17}{8}$

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C Fractions and percentages of amounts ●●○

5. Work out:

a) $\frac{3}{4}$ of 28 **b)** $\frac{2}{5}$ of 45 **c)** $\frac{5}{8}$ of \$72

d) $\frac{7}{10}$ of 250 **e)** $\frac{4}{9}$ of 63 **f)** $\frac{5}{6}$ of 3.6 kg

6. Work out:

a) 35% of 240 **b)** 7.5% of 800 **c)** 2.5% of \$1,600

d) 125% of 48 **e)** 0.5% of 3,000 **f)** 18% of 65 kg

D One amount as a percentage of another ●●○

⚠ **Watch out:** Put the part on top and the whole on the bottom, and make sure both are in the same units.

7. Answer each question.

a) What percentage of 250 is 45? **b)** What percentage of 64 is 24?

c) Write 18 out of 40 as a percentage. **d)** Write \$36 as a percentage of \$150.

e) Write 45 minutes as a percentage of 2 hours. **f)** Write 350 g as a percentage of 2 kg.

E Finding the whole amount ●●●

💡 **Remember:** Find 1% (or one part of the fraction) first, then build up to the whole.

8. Find the number.

a) 20% of a number is 14. **b)** $\frac{3}{4}$ of a number is 36.

c) 35% of a number is 91. **d)** $\frac{5}{8}$ of a number is 45.

F Problems ●●●

9. A netball team won 15 of its 24 games. What percentage of its games did it win?

10. A shop takes 30% off a \$90 pair of shoes. Find the sale price.

11. A 240 g bag of trail mix is $\frac{3}{8}$ nuts. How many grams are not nuts?

★ Extension

12. A price goes up by 20% and then comes down by 20%. What percentage of the original price is it now?

13. $\frac{2}{5}$ of a number is 36. What is 75% of the number?

14. A shop sells a jacket for $\frac{3}{4}$ of its usual price. Then it takes another 20% off. What fraction of the usual price is the final price?

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Fractions, Decimals and Percentages

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · convert *(panoni)*

A Converting ●○○

Remember: Per cent means "out of 100". To change a decimal to a percentage, multiply by 100.

1. Write each fraction as a decimal.

a) $\frac{1}{4}$ = _____ **b)** $\frac{3}{5}$ = _____ **c)** $\frac{5}{8}$ = _____ **d)** $\frac{7}{20}$ = _____

e) $\frac{3}{8}$ = _____ **f)** $\frac{11}{40}$ = _____ **g)** $\frac{23}{50}$ = _____ **h)** $\frac{3}{16}$ = _____

2. Write as a percentage.

a) 0.45 = _____ **b)** 0.03 = _____ **c)** 1.5 = _____ **d)** $\frac{4}{5}$ = _____

e) $\frac{9}{25}$ = _____ **f)** $\frac{3}{8}$ = _____ **g)** $\frac{11}{20}$ = _____ **h)** 0.006 = _____

3. Write as a fraction in simplest form.

a) 0.8 = _____ **b)** 0.35 = _____ **c)** 16% = _____ **d)** 62.5% = _____

e) 0.375 = _____ **f)** 0.08 = _____ **g)** 45% = _____ **h)** 7.5% = _____

B Ordering ●●○

Remember: Change every number to the same form, such as decimals, before you compare them.

4. Write each set in order, smallest first.

a) $0.7, \quad 68\%, \quad \frac{3}{4}$

b) $\frac{2}{3}, \quad 0.6, \quad 65\%$

c) $82\%, \quad \frac{4}{5}, \quad 0.85$

d) $1.4, \quad 135\%, \quad \frac{11}{8}$

e) $0.55, \quad 54\%, \quad \frac{5}{9}$

f) $1.9, \quad 185\%, \quad \frac{15}{8}$

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C Fractions and percentages of amounts ●●○

5. Work out:

a) $\frac{3}{4}$ of 36 **b)** $\frac{3}{5}$ of 35 **c)** $\frac{3}{8}$ of \$56

d) $\frac{9}{10}$ of 350 **e)** $\frac{5}{9}$ of 72 **f)** $\frac{2}{3}$ of 4.5 kg

6. Work out:

a) 45% of 260 **b)** 2.5% of 600 **c)** 7.5% of \$1,200

d) 150% of 36 **e)** 0.5% of 5,000 **f)** 24% of 75 kg

D One amount as a percentage of another ●●○

⚠ **Watch out:** Put the part on top and the whole on the bottom, and make sure both are in the same units.

7. Answer each question.

a) What percentage of 300 is 54? **b)** What percentage of 96 is 36?

c) Write 27 out of 60 as a percentage. **d)** Write \$42 as a percentage of \$175.

e) Write 36 minutes as a percentage of 3 hours. **f)** Write 450 g as a percentage of 3 kg.

E Finding the whole amount ●●●

💡 **Remember:** Find 1% (or one part of the fraction) first, then build up to the whole.

8. Find the number.

a) 40% of a number is 18. **b)** $\frac{2}{5}$ of a number is 26.

c) 15% of a number is 42. **d)** $\frac{3}{8}$ of a number is 27.

F Problems ●●●

9. A netball team won 21 of its 28 games. What percentage of its games did it win?

10. A shop takes 40% off an \$85 pair of shoes. Find the sale price.

11. A 320 g bag of trail mix is $\frac{5}{8}$ nuts. How many grams are not nuts?

★ Extension

12. A price goes up by 25% and then comes down by 25%. What percentage of the original price is it now?

13. $\frac{3}{5}$ of a number is 48. What is 25% of the number?

14. A shop sells a jacket for $\frac{4}{5}$ of its usual price. Then it takes another 25% off. What fraction of the usual price is the final price?

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